Pre-Calculus 11 - Reasoning and Analyzing Units 3+4 (V2)

Name : _____

Teacher's name: _____

Outcome	0	1	2	3	4	5	6	7
Reasoning and Analyzing								

1. For each of the following statements, identify if the statement is **always true**, **sometimes true**, or **never true**. Provide a brief justification.

   (a) If $-180° < \theta < 0°$, then $\sin\theta < 0$.

   (b) If $0° < \theta < 360°$ and $\sin\theta > 0$, then θ is in quadrant 1.

   (c) If $\dfrac{a}{k} > 0$, then $y = a(x-h)^2 + k$ has exactly two x-intercepts.

   (d) If the discriminant is equal to zero, then the vertex of the parabola lies on the x-axis.

   (e) If a quadratic function has a maximum value, then $k > 0$.

2. A student was asked to solve the equation and leave answers in exact form. Which step did he **first** make an error? Explain the error.

$$
\begin{aligned}
3(x-2)^2 &= 27 \\
(x-2)^2 &= 9 && \textbf{Step 1} \\
x-2 &= \pm 3 && \textbf{Step 2} \\
x &= 2 \pm 3 && \textbf{Step 3} \\
x &= \pm 5 && \textbf{Step 4}
\end{aligned}
$$

3. In each of the following, one item doesn't belong.

   (a) Which expression doesn't belong? Explain.

   i. $\sin 270^\circ$ ii. $\cos 180^\circ$ iii. $\tan 135^\circ$ iv. $\sin 90^\circ$

   (b) Which quadratic function doesn't belong? Explain.

   i. $y = -3(x+1)^2 - 4$ ii. $y = 2(x-1)^2 + 3$

   iii. $y = (x-4)^2 + 1$ iv. $y = 5x^2 + 2$